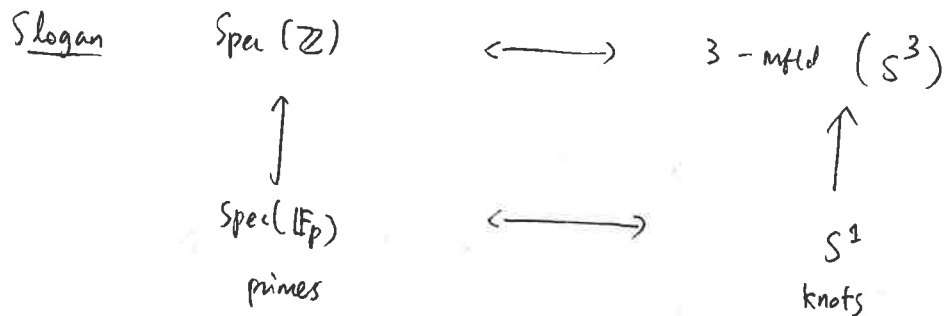


Artin - Verdier duality

Vaughan McDonald



Main Thm (Artin - Verdier) $X = \text{Spec } \mathcal{O}_K$, $K \mid \mathbb{Q}$, totally imaginary.

Then $\exists$ a nondegenerate pairing $H_{\text{et}}^r(X, F) \times \text{Ext}_X^{3-r}(F, \mathcal{L}_m) \rightarrow H^3(X, \mathcal{L}_m) \simeq \mathbb{Q}/\mathbb{Z}$
 for any constructible sheaf F .
 ($H_{\text{et}}^r(X, F)$, $\text{Ext}_X^{3-r}(F, \mathcal{L}_m)$ are finite)

Rmks i) The pairing is just the Yoneda product. since $H^r(X, F) \simeq \text{Ext}_X^r(\mathbb{Z}_X, F)$.

ii) If X not totally imaginary, issues w/ $\mathbb{Z}$ -torsion.

iii) No assumptions on the size of stacks of F !

iv) version w/ $U \subset X$ dense ^{open} subscheme (think $\text{Spec}(\mathcal{O}_{K,S})$, S finite set of places)

v) The famous "Poitou - Tate 9-term sequence" follows from applying $\text{Ext}(F, -)$ to $\text{Spec}(\mathcal{O}_{K,S})$

$$0 \rightarrow K^\times \rightarrow A_K^\times \rightarrow A_K^\times / K^\times \rightarrow 0.] \text{ } S\text{-integral version}$$

+ applying global CFT and Artin - Verdier duality.

$$\begin{array}{ccc}
 \text{vi)} & K \longrightarrow & \mathbb{F}_q(\pi) \\
 & & \downarrow \\
 & & 3\text{-dim'l} \\
 & & \parallel \\
 & & 2+1
 \end{array}$$

Proof strategy. First need to show $\textcircled{A}$ $H^3(X, \mathcal{G}_m) \simeq \mathbb{A}/\mathbb{Z}$ + finiteness of H^r, Ext^r $\textcircled{B}$

$$\Rightarrow m^r: H_{\text{et}}^r(X, F) \rightarrow (\text{Ext}_X^{3-r}(F, \mathcal{G}_m))^v$$

$\textcircled{C}$ |
prove m^r is an isomorphism by induction on r .

We'll use the sequence $0 \rightarrow \mathcal{G}_m \rightarrow j_* \mathcal{G}_m|_{\text{Spec } k} \rightarrow \coprod_{\substack{p \in X \\ \text{closed pt}}} \mathbb{Z}/p \rightarrow 0$

$$j: \text{Spec } k \rightarrow \text{Spec } \mathcal{O}_K$$

$$(0 \rightarrow \mathcal{O}_K^X \rightarrow K^X \rightarrow K^X/\mathcal{O}_K^X \rightarrow 0)$$

LES $0 \rightarrow \mathcal{O}_K^X \rightarrow K^X \rightarrow \bigoplus_p \mathbb{Z} \rightarrow$

$$\rightarrow \text{Pic}(X) \rightarrow H^1(X, j_* \mathcal{G}_m) \rightarrow H^1(X, \bigoplus \mathbb{Z}/p)$$

$$\rightarrow H^2(X, \mathcal{G}_m)$$

Claim. $H^q(X, \bigoplus \mathbb{Z}/p) = \begin{cases} 0, & q=1, q \geq 2 \\ \bigoplus \mathbb{A}/\mathbb{Z}, & q=2 \end{cases}$

$$= \bigoplus_p H^q(\text{Spec } \mathbb{F}_p, \mathbb{Z})$$

$$H^1(\text{Spec } \mathbb{F}_p, \mathbb{Z}) = \text{Hom}(\hat{\mathbb{Z}}, \mathbb{Z}) = 0.$$

$\text{Spec } \mathbb{F}_p$ has cohomological dim 1 on torsion objects

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{A} \rightarrow \mathbb{A}/\mathbb{Z} \rightarrow 0 \Rightarrow H^q(\text{Spec } \mathbb{F}_p, \mathbb{Z}) \simeq H^{q-1}(\text{Spec } \mathbb{F}_p, \mathbb{A}/\mathbb{Z})$$

Claim. $R^q j_* (\mathcal{G}_m, X) = 0$ for $q > 0$.

Leray SS gives $H^i(X; j_* \mathcal{G}_m, X) \simeq H^i(X, \mathcal{G}_m, X) \simeq H^i(\text{Gal}(\bar{K}/K), \bar{K}^*)$

$$0 \rightarrow H^2(X, \mathcal{G}_m) \rightarrow \text{Br } K \rightarrow \bigoplus_p \mathbb{Q}/\mathbb{Z} \rightarrow H^3(X, \mathcal{G}_m) \rightarrow H^3(\text{Gal}(\bar{K}/K), \bar{K}^*)$$

! CFT

CFT $\Rightarrow$ $0 \rightarrow \text{Br } K \rightarrow \bigoplus_p \mathbb{Q}/\mathbb{Z} \xrightarrow{\text{sum}} \mathbb{Q}/\mathbb{Z} \rightarrow 0$

In summary, $H^i(X, \mathcal{G}_m) = \begin{cases} \mathcal{O}_K^* & q=0 \\ \text{Pic } X & q=1 \\ 0 & q=2, q>3 \\ \mathbb{Q}/\mathbb{Z} & q=3 \end{cases}$

The exact seq. $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$ hit up $\text{Ext}(-, \mathcal{G}_m)$

$$\Rightarrow \text{Ext}_X^q(\mathbb{Z}/n, \mathcal{G}_m) = \begin{cases} H^1(\mathcal{O}_K), & q=0 \\ \text{Pic } X / n & q=1 \\ \mathbb{Z}/n & q=2 \\ 0 & q>2 \end{cases}$$

an ext'n of

$$\text{Pic } X[n] \text{ by } q=1$$

$$\mathcal{O}_K^* / (\mathcal{O}_K^*)^n$$

ⓑ Finitude + vanishing in degrees $q > 3$.

Many reductions: Call a sheaf F punctual if it is of the form $i_{*} E \simeq F$.

for $i: V \rightarrow X$ proper closed subset.

If punctual, $H^r(X, \mathbb{Z}_* E) \cong H^r(V, E) \Rightarrow$ finiteness $\checkmark$.

Likewise for Ext^r .

Any constructible sheaf differs from a sheaf of the form $j_! C$ for $j: U \rightarrow X$ dense open $\xleftarrow{\text{lisse}}$ up to a punctual sheaf. $\Rightarrow$ reduce to the case of a constant sheaf, up to replacing X by U open.

By further shrinking U , relate to Galois cohomology.

③ standard pull-push argument let you assume $F = \mathbb{Z}/p$, and $X = \text{Spec } \mathcal{O}_K$, $\mathcal{O}_K \supset \mathbb{F}_p$.

Another standard (effaceability) arg:

If m^r isom is. for all $r < r_0$ and all F , then m^{r_0} is injective.

All this amounts to is proving $\# H^r(X, \mathbb{Z}/p) \geq \# \text{Ext}^{3-r}(\mathbb{Z}/p, G_m)$.

$r = 0, 1, 3$ easy. $r = 2$ real work, relies on nondegeneracy of the local Hilbert symbol.